

UTHSAV CHITRA

Department of Computer Science, Johns Hopkins University

<https://uthsavc.github.io>

ACADEMIC POSITIONS

Johns Hopkins University

2025-present

Assistant Professor, Department of Computer Science
Member, Data Science and AI Institute
Member, Center for Computational Biology

EDUCATION AND TRAINING

Broad Institute of MIT and Harvard

2024-2025

Postdoctoral fellow, Eric and Wendy Schmidt Center

Princeton University, Princeton, New Jersey

2018-2024

Ph.D., Computer Science

Brown University, Providence, Rhode Island

2013-2017

Sc.B. Mathematics, A.B. Computer Science, A.B. Applied Math

GPA: 4.0/4.0

HONORS AND AWARDS

Supervisor of the Year Nominee, JHU University Experiential Learning

2026

Best Overall Poster, Eric and Wendy Schmidt Center Symposium, Broad Institute of MIT and Harvard

2025

Rising Stars in Data Science, UChicago/UC San Diego/Stanford Data Science Institutes

2024

Best Paper Award, RECOMB Satellite Workshop on Computational Cancer Biology

2024

Siebel Scholarship

2022

- Award of \$35,000 given annually to 85 top graduate students worldwide in computer science, bioengineering, and business.

Best Reviewer Award, International Conference on Machine Learning (ICML)

2021, 2022

NSF Graduate Research Fellowship

2020

Jerome Stein Memorial Award, Brown University Applied Math Department

2017

- Given to the top two students who “show outstanding potential in an interdisciplinary area that involves applied mathematics.”

Phi Beta Kappa, Brown University (elected junior year, top 2% of class)

2016

Top 200, William Lowell Putnam Math Competition

2015

First Place, Brown University Hartshorn-Hypatia Math Prize Exam

2013

Semi-finalist, Siemens Competition (research project in number theory)

2012

USA Junior Math Olympiad qualifier

2011

PUBLICATIONS

* denotes joint first authorship, † denotes joint senior authorship. Underline indicates (co-)mentored trainee.

Mapping spatial gradients in spatial transcriptomics data with score matching. An Wang, Donald Geman, **Uthsav Chitra**[†], Laurent Younes[†]. *In preparation*.

Multi-GASTON Learns Tissue-Intrinsic Coordinates Across Multiple Spatial-omics Slices. Clover Zheng, Cong Ma[†], **Uthsav Chitra**[†], Benjamin J. Raphael[†]. *In preparation*.

CRIMP efficiently quantifies subcellular gene expression patterns from image-based spatial transcriptomics. Claire Wu, Hannah Schlueter, **Uthsav Chitra**[†], Caroline Uhler[†]. *In preparation*.

Decoding the causal drivers of spatial cellular topology. Prannav Shankar, Huan Liang, **Uthsav Chitra**[†], Rohit Singh[†]. *iScience* (2026). Accepted to *RECOMB-seq* 2025.

Anomaly Detection in Spatial Transcriptomics via Spatially Localized Density Comparison. Gary Hu, Julian Gold, **Uthsav Chitra**, Sunay Joshi, Benjamin J. Raphael. *Bioinformatics* (2025). Accepted to *ISMB 2025*.

GASTON-Mix: a unified model of spatial gradients and domains using spatial mixture-of-experts. **Uthsav Chitra**, Shu Dan, Fenna Krienen, Benjamin J. Raphael. *Bioinformatics* (2025). Accepted to *ISMB 2025*.

Spatial metabolic gradients in the liver and small intestine. Laith Samarah, Clover Zheng, Xi Xing, Won Dong Lee, Amichay Afriat, **Uthsav Chitra**, Michael MacArthur, Wenyun Lu, Connor Jankowski, Cong Ma, Craig Hunter, Benjamin J. Raphael, Joshua Rabinowitz. *Nature* (2025).

Mapping the topography of spatial gene expression with interpretable deep learning. **Uthsav Chitra**, Brian J. Arnold, Hirak Sarkar, Cong Ma, Sereno Lopez-Darwin, Kohei Sanno, Benjamin J. Raphael. *Nature Methods* (2025). Accepted to *RECOMB 2024*.

Resolving discrepancies between chimeric and multiplicative measures of higher-order epistasis. **Uthsav Chitra***, Brian J. Arnold*, Benjamin J. Raphael. *Nature Communications* (2025).

A latent variable model for evaluating mutual exclusivity between driver mutations in cancer. Ahmed Shuaibi*, **Uthsav Chitra***, Benjamin J. Raphael. In preparation. *RECOMB Satellite Workshop on Computational Cancer Biology (RECOMB-CCB)*, 2024. **Best Paper Award**.

A count-based model for delineating cell-cell interactions in spatial transcriptomics data. Hirak Sarkar*, **Uthsav Chitra***, Julian Gold, Benjamin J. Raphael. *Bioinformatics* (2024). Accepted to *ISMB 2024*.

Belayer: Modeling discrete and continuous spatial variation in gene expression from spatially resolved transcriptomics. Cong Ma*, **Uthsav Chitra***, Shirley Zhang, Benjamin J. Raphael. *Cell Systems* (2022). Accepted to *RECOMB 2022*.

NetMix2: Unifying network propagation and altered subnetworks. **Uthsav Chitra***, Tae Yoon Park*, Benjamin J. Raphael. *Journal of Computational Biology* (2022). Accepted to *RECOMB 2022*.

Quantifying and Reducing Bias in Maximum Likelihood Estimation of Structured Anomalies. **Uthsav Chitra**, Kimberly Ding, Jasper C. H. Lee, Benjamin J. Raphael. *International Conference on Machine Learning (ICML)* 2021.

NetMix: A network-structured mixture model for reduced-bias estimation of altered subnetworks. Matthew A Reyna*, **Uthsav Chitra***, Rebecca Elyanow, Benjamin J. Raphael. *Journal of Computational Biology* (2021). Accepted to *RECOMB 2020*.

Analyzing the Impact of Filter Bubbles on Social Network Polarization. **Uthsav Chitra** and Christopher Musco. *ACM International Web Search and Data Mining Conference (WSDM)* 2020. Also appeared at KDD WISDOM 2019 workshop.

Random Walks on Hypergraphs with Edge-Dependent Vertex Weights. **Uthsav Chitra** and Benjamin J. Raphael. *International Conference on Machine Learning (ICML)* 2019.

Committee Selection is More Similar Than You Think: Evidence from Avalanche and Stellar. Tarun Chitra and **Uthsav Chitra**. *Manuscript*, 2019.

TEACHING

Johns Hopkins University

Instructor, EN 601.752: Advanced Topics In Single-cell and Spatial Genomics

Spring 2026

Princeton University

Instructor/Curriculum Developer, Princeton Prison Teaching Initiative

2019-2023

- Taught college-accredited math classes at NJ state prisons. Developed and taught first-ever Java programming course for NJ state prisons.

Brown University

Teaching Assistant, MATH 1560: Number Theory

Spring 2016, Spring 2017

Teaching Assistant , CSCI 1570: Design and Analysis of Algorithms	<i>Fall 2015, Fall 2016</i>
Teaching Assistant , CSCI 1450: Probability in Computing	<i>Spring 2015</i>
Teaching Assistant , CSCI 0530: Linear Algebra for CS	<i>Fall 2014</i>
Grader , MATH 1530: Abstract Algebra	<i>Spring 2014</i>
<u>Miscellaneous</u>	
Guest Lecturer , BME590-7595 - Spatial-Omics: Methods and Analysis, Duke University	<i>Spring 2026</i>
Counselor , Program in Mathematics for Young Scientists (PROMYS)	<i>Summer 2014</i>
Teaching Assistant , Art of Problem Solving	<i>2012-2016</i>

TALKS / POSTERS

Modeling the geometry of spatial gene expression

ECE seminar, Johns Hopkins University	<i>October 2025</i>
Biostatistics seminar, Johns Hopkins University	<i>September 2025</i>

GASTON-Mix: a unified model of spatial gradients and domains using spatial mixture-of-experts

Intelligent Systems for Molecular Biology (ISMB/ECCB)	<i>July 2025</i>
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Machine learning for spatial and network biology

Computational biology seminar, Carnegie Mellon University	<i>March 2025</i>
Computer science/BME seminar, Johns Hopkins University	<i>March 2025</i>
Computer science seminar, University of Maryland	<i>February 2025</i>
Rising Stars in Data Science, UC San Diego	<i>November 2024</i>

Modeling spatial gene expression with complex analysis and deep learning

Computational and Systems Biology (CSB) seminar, MIT	<i>November 2024</i>
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Mapping the topography of spatial gene expression with interpretable deep learning.

Eric and Wendy Schmidt Center Symposium, Broad Institute of MIT and Harvard (poster)	<i>April 2025</i>
Models, Inference, & Algorithms seminar, Broad Institute of MIT and Harvard	<i>March 2025</i>
Conference on Research in Computational Molecular Biology (RECOMB)	<i>May 2024</i>
Single Cell Analyses, Cold Spring Harbor Laboratory (poster)	<i>November 2023</i>
Rutgers-Princeton Cancer Research Symposium (poster)	<i>October 2023</i>
NCI Junior Investigator (JI) Annual Meeting	<i>August 2023</i>

Belayer: Modeling discrete and continuous spatial variation in gene expression from spatially resolved transcriptomics

Wang Lab Meeting, Broad Institute	<i>July 2023</i>
NCI Spring School on Algorithmic Cancer Biology	<i>March 2023</i>

Algorithms for understanding the spatial and network organization of biological systems

Chen Lab, Broad Institute	<i>July 2024</i>
Campbell Lab, UToronto	<i>April 2024</i>
Final Public Oral (FPO, i.e. thesis defense), Princeton University	<i>March 2024</i>
Knowles/Azizi Lab, Columbia University	<i>September 2023</i>
Herbert Irving Comprehensive Cancer Center, Columbia University	<i>September 2023</i>
Pe'er Lab, MSKCC	<i>August 2023</i>

Modeling spatial variation in gene expression and copy number aberrations

Brigham Women's Hospital Advanced Biomedical Computation Series	<i>March 2023</i>
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Leveraging network and spatial structure to model high-dimensional biological data

Sankararaman/Pimentel Labs, UCLA	<i>April 2023</i>
Pe'er Lab, Columbia	<i>April 2023</i>
Hormoz Lab, DFCI Data Science	<i>February 2022</i>

NetMix2: Unifying network propagation and altered subnetworks	
Conference on Research in Computational Molecular Biology (RECOMB)	<i>May 2022</i>
Quantifying and Reducing Bias in Maximum Likelihood Estimation of Structured Anomalies	
International Conference on Machine Learning (ICML)	<i>July 2021</i>
NetMix: A network-structured mixture model for reduced-bias estimation of altered subnetworks	
Conference on Research in Computational Molecular Biology (RECOMB)	<i>June 2020</i>
Algorithms for Analyzing Networks with Vertex Weights	
Princeton University Generals Exam	<i>May 2020</i>
Analyzing the Impact of Filter Bubbles on Social Network Polarization	
ACM International Web Search and Data Mining Conference (WSDM)	<i>February 2020</i>
KDD WISDOM Workshop	<i>August 2019</i>
Random Walks on Hypergraphs with Edge-Dependent Vertex Weights	
SIAM Conference on Discrete Mathematics	<i>June 2022</i>
International Conference of Machine Learning (ICML)	<i>June 2019</i>

ADVISING

Postdoctoral Fellows

James O'Quinn, JHU CS	<i>2026-present</i>
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PhD students

Leyla Rasheed, JHU CS	<i>2026-present</i>
Luiz Mata Lopez, JHU CS	<i>2026-present</i>
Thu Nga (Nerissa) Vu, JHU ChemBE (co-advised with Sangkyun Cho)	<i>2025-present</i>
Sayuni Dharmasena, JHU CS	<i>2025-present</i>

Master's students

Alice Mao, JHU Bioinformatics	<i>2025-present</i>
Dhanav Handa, JHU CS	<i>2025-present</i>

Undergraduate students

Harry Jiang, JHU CS (with Hongkai Ji)	<i>2026-present</i>
Mischa Kumar, JHU CS	<i>2026</i>
Chelsea Wong, JHU CS	<i>2026</i>
Kevin He, JHU AMS/Economics	<i>2026</i>
Aditya Jain, JHU BME/AMS/CS	<i>2026-present</i>
• Received Pistritto Fellowship to support research.	
Angela Ning, JHU CS/Biology	<i>2025-present</i>
Evan Li, JHU CS	<i>2025-present</i>
Claire Wu, MIT undergraduate	<i>2024-present</i>
• Won Best Poster at the 2025 MIT Biological Engineering Undergraduate Research Symposium.	
Tanvi Haldiya, Princeton CS	<i>Fall 2023</i>
Jairam Hathwar, Princeton CS	<i>Fall 2023</i>
Kohei Sanno, Princeton CS. <i>Next position: UC Berkeley CS PhD student.</i>	<i>2023-2024</i>
Sunay Joshi, Princeton Math. <i>Next position: UPenn Applied Math PhD.</i>	<i>2022-2024</i>
Kimberly Ding, Princeton CS. <i>Next position: Quant at Virtu Financial.</i>	<i>2019-2021</i>
• Received an “ Outstanding Computer Science Senior Thesis Prize ”	
Shirley Zhang, Princeton CS. <i>Next position: Harvard CS PhD.</i>	<i>2020-2022</i>
• Received an NSF Graduate Research Fellowship	

Co-mentored PhD students

An Wang, JHU AMS (advisors: Laurent Younes, Donald Geman)	<i>2025-present</i>
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Clover Zheng, Princeton CS (advisor: Ben Raphael)	2022-present
Ahmed Shuaibi, Princeton QCB (advisor: Ben Raphael). <i>Now Research Scientist at Meta.</i>	2020-2025
• Won Best Paper Award at RECOMB-CCB workshop.	
Madelyne Xiao, Princeton CS (advisor: Ben Raphael)	2022

REVIEWING

Conferences

RECOMB 2023-2026, ISMB 2023-2026, ICML 2021-2023 (**Top 10% Reviewer**, 2021-2022), NeurIPS 2021, ICML 2024 AccMLBio workshop, RECOMB 2020 poster session

Journal Reviewing

Nature Biotechnology (x2), *Nature Methods*, *PNAS*, *Genome Biology*, *JASA*, *Bioinformatics*, *PLOS Computational Biology*, *Communications Biology*, *Bioinformatics Advances*, *Frontiers in Big Data*, *Computational and Structural Biotechnology Journal*.

SERVICE

To Johns Hopkins University

CS Master's Admissions Reviewer	2026
DSAI Postdoctoral Fellowship Selection Committee	2026
Internal reviewer, Apple Scholars in AI/ML PhD Fellowship	2025
JHU CS Faculty Research Panel	2025

To the Profession

Area Chair: ACM BCB 2025.
Program Committee: ISMB 2025, ISMB 2026, ECCB 2026.

To the Greater Community

Visiting Fellowship in Assessment, College Board Advanced Placement (AP) program	2025
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Thesis Committee

Gus Fridell, JHU Biology

Graduate Board Oral (GBO) / Doctoral Board Oral (DBO) Committee

Julie Xian, JHU BME	2026
Edward Lui, JHU CS	2026
Hyun Joo (Hayden) Ji, JHU CS	2026
Celine Hoh, JHU CS	2025
April Kim, JHU CS	2025

Graduate/Undergraduate Student Committees

Member, Princeton COS Graduate Student Committee	2022-2023
Member, Princeton Graduate Engineering Council	2021-2023
Member, Princeton COS Ad Hoc Committee	2021
Officer, Brown Math Departmental Undergraduate Group	2015-2017
Mentor, Brown Matched Advising Program for Sophomores	2016-2017

INDUSTRY EXPERIENCE

Software Engineer, Facebook

2017-2018

- Built infrastructure, machine learning models, and data pipelines for improving ad quality.

Software Engineering Intern, Facebook

Summer 2016

- Reduced upload time for video ads by 20%.

Hobbies/interests: **Bouldering**, biking, crosswords and other puzzles.